CSCI 652 Distributed Systems

Term Paper Instructions and Topics

Choose **one** topic and let me know via email by September 25, 2026. Term papers must be written by individuals – this is not group work. Maximum number of persons per broad topic is 3 (email time stamp will be used). Individuals writing term paper on the same broad topic will each focus on different aspects. A presentation in class will precede term paper submission. Individuals should present and write their term paper independently.
A scribe will be assigned for each presentation; one scribing assignment per student. Scribe tasks: 1. Write a summary of the presentation - challenges, ideas, methodologies, and results presented. 2. Record questions and responses.

We will discuss each topic in class. **All students regardless of their topic are expected to actively participate – listen and learn, ask questions and provide feedback.** Quizzes 4 and 5 will include questions related to term paper presentations.

- Fault-tolerant Distributed Computing
- Security and Privacy in Distributed Computing
- Distributed Caching
- Distributed Machine Learning
- Federated learning
- Service Oriented Architectures
- Serverless Computing
- Cyber Physical Systems, Internet of Things, Sensors
- Pervasive Computing
- Distributed Computing on the Edge
- Distributed Agentic AI
- LLMs and Distributed Computing
- Drones, Unmanned Vehicles, UAVs
- Block Chain and Smart Contracts
- Classic Paxos and Fast Paxos
- RAFT algorithm

Please note this is a course on Distributed Systems. The term paper should be focused on one or more issues related to distributed systems.
Please consult with me if you wish to take up a topic different from those listed above.

Use of AI tools should be limited to understanding published content. Presentation slides and term papers MUST be prepared and written by the concerned student. Presenters should be able to respond to questions related to the contents in their submitted slides and reports.

Timeline:

1. September 25, 2026 - Choose topic.
2. October 08, 2026 - Identify at least 3 published articles on the topic for reference and study. They should all be quality articles published in reputed venues. You may use generative AI tools to choose most appropriate articles. Choosing appropriate articles of substance is very important. We will maintain a google sheet for keeping track of who is reading which paper. I will comment on choice of your papers by October 15, 2026.
3. Instructions for preparing slides and writing the term paper will be available.
4. Presentations – 25 Minutes presentation (typically 2 per class) will begin from November 3, 2026; Draft slides should be shared with me - **by preceding Friday for Tuesday presentations and Monday for Thursday presentations**. I will provide comments/feedback.
   Do not use Generative AI tools to prepare your slides.
5. Optional – I will provide comments on term paper drafts submitted before November 30, 2026.
   Draft submissions will **not** be graded; it is to your benefit to submit early to receive feedback and submit a better-quality term paper.
   Do not use Generative AI tools to prepare your term paper.
6. December 10, 2026 – Submit term paper for grading.

Presentation in Class

Presentation time will be 25 minutes; plan to finish within this time. Presentation will be followed by a 10-minute Q and A session. All other students are expected to ask questions, provide feedback and make constructive comments. This will be part of class participation. Questions pertaining to class presentations will be included for quizzes Q4 and Q5.

Quiz 4 - term paper presentations on 11/3, 11/5, 11/10 and 11/12.

Quiz 5 - term paper presentations on 11/17,11/19, 12/1 and 12/3.

Instructions for preparing Presentation Slides

Prepare 10-15 slides; 1 for intro, 1 for conclusions and 1 for references.

Use remaining slides to give an overview of the topic, ongoing work; what you have learnt. Make constructive comments and/or list open problems/challenges.

Figures, sketches, graphs, plots etc. convey the message better than words in oral presentations.

Instructions for Writing your Term Paper

Page Limit: 6-8 pages in one-column format; font size 11.

Content: Title; Abstract (6-8 lines); Introduction (1 page); distributed algorithms and architectures, examples and software/hardware (1-2 pages); research advances, developments and applications (2-3 pages); scope, critic and challenges (1 to 2 pages); conclusion (6-8 lines); and References (3- 5 articles).

Detailed Instructions –

The first step to writing a good paper is to read published articles. Read as many as you can. Read at least 3 selected articles thoroughly.

Writing a technical paper is not a simple task. One motivating factor is learning how to write a good technical paper. Treat this is as an opportunity to learn to write technical papers, not just another task to get a grade in a course. Paper writing is challenging and will improve with practice. If you make a start and pay attention to details, you will enjoy writing. When reading papers, you MUST pay attention to how a paper (e.g., a paper you are reading) is written – language, grammar, articulation of contributions and results, presentation of figures, tables and algorithms.

*Remember someone else is reading your paper - you should put yourself in the position of the reader when you proofread your paper. Proofreading what you write is critical. Unless the reader can understand, you cannot expect them to appreciate your knowledge.

Simple tips -

1. Use simple English.
2. Always remember you are writing for someone else to read (just like any producer who makes things for consumers) –quality is important.
3. Do not make spelling errors – do not depend on online spell checks completely. For example, 'remainder' and 'reminder' are both spelt right, but they have different meanings.
4. Each sentence should be approximately 1 to 1.5 lines.
5. Minimize short sentences (less than half a line).
6. Minimize long sentences (more than 2 lines).
7. Each sentence should be grammatically correct. There should be a smooth flow from one sentence to the next.
8. Know the difference between 'it's' and 'its' – 'it's' is short for 'it is'.
9. Do not overuse adjectives and adverbs.
10. Know when to use the article 'the' – pay attention to the use of 'the' here.
11. Proofread each sentence in the context of the text around it – there should be a smooth flow.
12. The space available to write is not unlimited. Just like any of us, the reader does not like to read crap. Every sentence should be well crafted and written with the reader in mind.
13. More ……

References, Citing and Copyright

The following are very important.

1. DO NOT copy and paste from another source (papers, textbooks, Internet sources, another student's written material, lecture notes, slides, etc.).
    a. If you cannot change authors statements (proofs, lemmas etc.); place such statements in quotes. But there should be no more than 2 such

instances in the paper.

2. We must acknowledge other's work (including Gen AI tools). We must cite appropriate references. When you read papers, pay attention to how authors cite other authors' work.

3. This is your term paper. You should write it (not AI). Every sentence you write in the paper should be your own. Read and understand what others have written, then write what you have understood on your own.
4. Copy and paste is tantamount to stealing in academics. Do NOT submit copied material for grading. If you do, this it will be treated as case of "Academic Integrity Violation".
5. This term paper is not your original paper (meaning you are reporting your original work). Most of the material will be from the three articles you have read, but the writing in the term paper must be yours.

Specific to Term Paper

Title: You may have a more specific title to reflect the content covered in the papers you read. The length of the title should be 10 to15 words in length. The title should attract the attention of the reader and at the same time it should not be misleading.

Abstract: The abstract of your term paper is an **abstract** of the contents in the rest of the paper**.** The abstract should be comprehensible to a layperson in computer science. Anyone who reads the abstract should be able to visualize what is in the rest of the paper at a high level. The abstract should be self-contained.

Introduction: This section introduces the reader to the –

- topics discussed in the term paper;
- problems addressed; and
- term paper itself.

Literature Review: Compare and contrast the work presented in the research articles you have read.

Critique, Scope and Discussion: This is one section where your comprehension and creativity can be demonstrated. Here, you can criticize the work presented in the three papers. You are free to compare contributions in the three papers and beyond. Your criticism should be justified. For example, if you say '*scheme A is not scalable*', you should back up your statement with justification. You can use more than 2 pages if you need to – however, every sentence should be carefully crafted.

Conclusion: Conclude the term paper in your own words. This is not a summary of the conclusions in papers you have read.

References: Provide complete citation for each of the references (3 papers of focus plus others (optional)).

Instructions ***for Reading Articles:***

The first step to writing a paper is to read published articles. Read as many as you can. Read at least three selected articles thoroughly.

After reading each paper (perhaps more than once), you should be able to answer the following questions:

1. What is the main contribution (research or development) of the work presented?
2. Is the technical contribution unique?
3. What are the strengths and weaknesses of the work presented in the articles?
4. Can you criticize the work presented?
5. Pay attention to the style of writing, organization, presentation of results, etc.